

## Phase 8: Data Management

### Project: SmartCare CRM (Healthcare Management System)

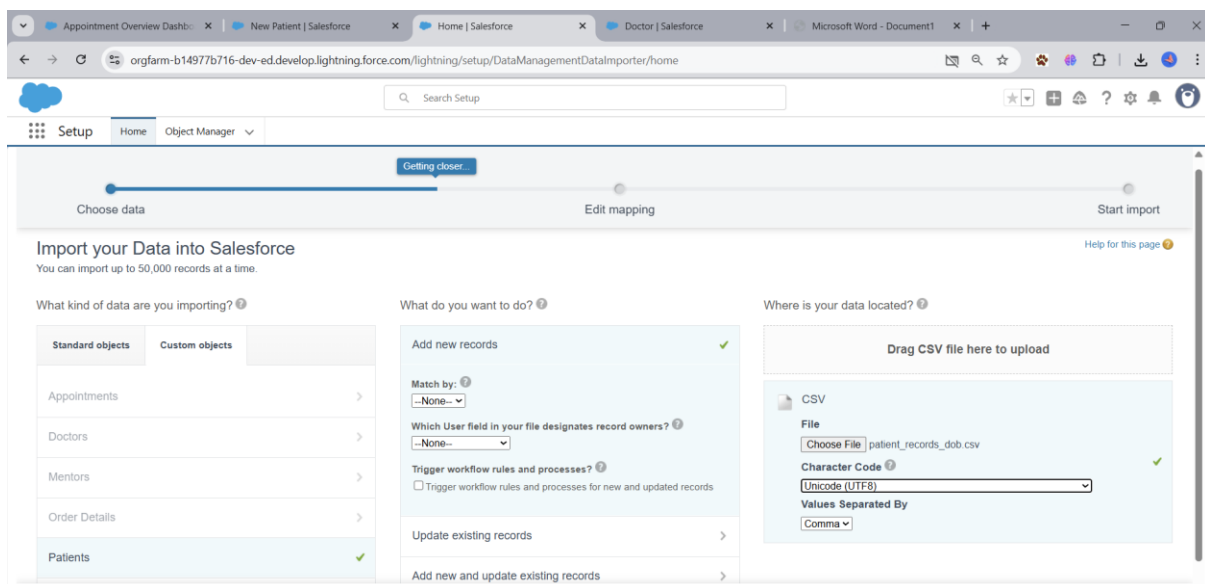
#### Purpose:

This phase focuses on ensuring data quality and security within the healthcare system. Key administrative tasks were performed to import patient records in bulk, prevent duplicate entries, and secure the data through regular backups.

#### Data Import using the Data Import Wizard

**Goal:** To efficiently add multiple new Patient records into the application from an external CSV file, demonstrating how the system can be populated with data in bulk.

**Implementation:** A CSV file containing 500 patient records was prepared, with fields including Name, Date of Birth, Gender, Email, Contact Number, Blood Group, Emergency Contact, and Medical History. The Data Import Wizard was launched, and the Patient custom object was selected for the import.



The screenshot displays the Salesforce Data Import Wizard interface. The browser address bar shows the URL: `orgfarm-b14977b716-dev-ed.develop.lightning.force.com/lightning/setup/DataManagementDataImporter/home`. The interface includes a top navigation bar with tabs for Appointment Overview Dashboard, New Patient | Salesforce, Home | Salesforce, Doctor | Salesforce, and Microsoft Word - Document1. Below the navigation bar is a search bar labeled "Search Setup". The main content area is titled "Import your Data into Salesforce" and includes a progress bar with three steps: "Choose data", "Edit mapping", and "Start import". The "Choose data" step is currently active. The interface is divided into three columns. The left column, "What kind of data are you importing?", shows a list of standard objects (Appointments, Doctors, Mentors, Order Details, Patients) and a "Custom objects" tab. The "Patients" object is selected. The middle column, "What do you want to do?", shows options for "Add new records", "Update existing records", and "Add new and update existing records". The "Add new records" option is selected. The right column, "Where is your data located?", shows a "Drag CSV file here to upload" section. A CSV file named "patient\_records\_dob.csv" is uploaded, and the "Character Code" is set to "Unicode (UTF8)".

#### Bulk Data Load Jobs

After uploading the file and completing the mapping process, the import job was submitted. Once completed, its status was verified under the "Bulk Data Load Jobs" page, confirming that all new records had been successfully inserted. The new patients were then visible under the Patients tab in the SmartCare CRM app.

The screenshot shows the Salesforce Bulk Data Load Jobs page. The left sidebar has a search bar with 'data im' and a 'Data Import Wizard' link. The main content area is titled 'Bulk Data Load Jobs' and shows details for a job with ID '750gK00000Dsc8r'. The job is submitted by 'Hospital Admin', started on 9/26/2025 at 5:07 AM PST, and ended at the same time. It is a 'Bulk V1' job of type 'Insert'. The status is 'Closed'. The table below shows the job's progress: 756 total records, 428 API active, 0 apex active, 100% progress, 500 records processed, 49 records failed, and 0 retries.

| Job ID                      | Submitted By           | Job Type            | Bulk V1 | Status                          | Closed |
|-----------------------------|------------------------|---------------------|---------|---------------------------------|--------|
| 750gK00000Dsc8r             | Hospital Admin         | Operation           | Insert  | Total Processing Time (ms)      | 756    |
| Start Time                  | 9/26/2025, 5:07 AM PST | Queued Batches      | 0       | API Active Processing Time (ms) | 428    |
| End Time                    | 9/26/2025, 5:07 AM PST | In Progress Batches | 0       | Apex Processing Time (ms)       | 0      |
| Time to Complete (hh:mm:ss) | 00:05                  | Completed Batches   | 1       | Failed Batches                  | 0      |
| Object                      | Patient                | Progress            | 100%    | Records Processed               | 500    |
| External ID Field           |                        | Records Failed      | 49      | Retries                         | 0      |
| Content Type                | CSV                    |                     |         |                                 |        |
| Concurrency Mode            | Parallel               |                     |         |                                 |        |
| API Version                 | 64.0                   |                     |         |                                 |        |

| View Request                 | View Result                 | Batch ID        | Start Time         | End Time           | Total Processing Time (ms) | API Active Processing Time (ms) | Apex Processing Time (ms) | Records Processed | Records Failed | Retry Count | State Message | Status    |
|------------------------------|-----------------------------|-----------------|--------------------|--------------------|----------------------------|---------------------------------|---------------------------|-------------------|----------------|-------------|---------------|-----------|
| <a href="#">View Request</a> | <a href="#">View Result</a> | 751gK00000BEsJx | 9/26/2025, 5:07 AM | 9/26/2025, 5:07 AM | 756                        | 428                             | 0                         | 500               | 49             | 0           |               | Completed |

## Data Backup using the Data Export Service

**Goal:** To establish a regular data backup strategy, ensuring that patient data is always protected and recoverable.

**Implementation:** The Salesforce Data Export service was used to schedule a weekly export of all healthcare data, including Patients, Appointments, and Doctors. This provides administrators with downloadable backup files for offline storage and recovery purposes.

The screenshot shows the Salesforce Data Export 'Schedule Data Export' page. The left sidebar has a search bar with 'data ex' and a 'Data Export' link. The main content area is titled 'Schedule Data Export' and shows settings for a scheduled export. The export file encoding is set to 'Unicode (UTF-8)'. The frequency is set to 'On day 1 of every month'. The start date is 9/26/2025 and the end date is 10/26/2025. The preferred start time is 1:00 AM. The page includes 'Save' and 'Cancel' buttons.

## Conclusion:

Through this phase, essential data management processes were successfully implemented. The Patient object was populated with bulk data using the Import Wizard, duplicate records were prevented with validation rules, and a reliable data export strategy was established. This strengthens the overall reliability and security of the healthcare management application.